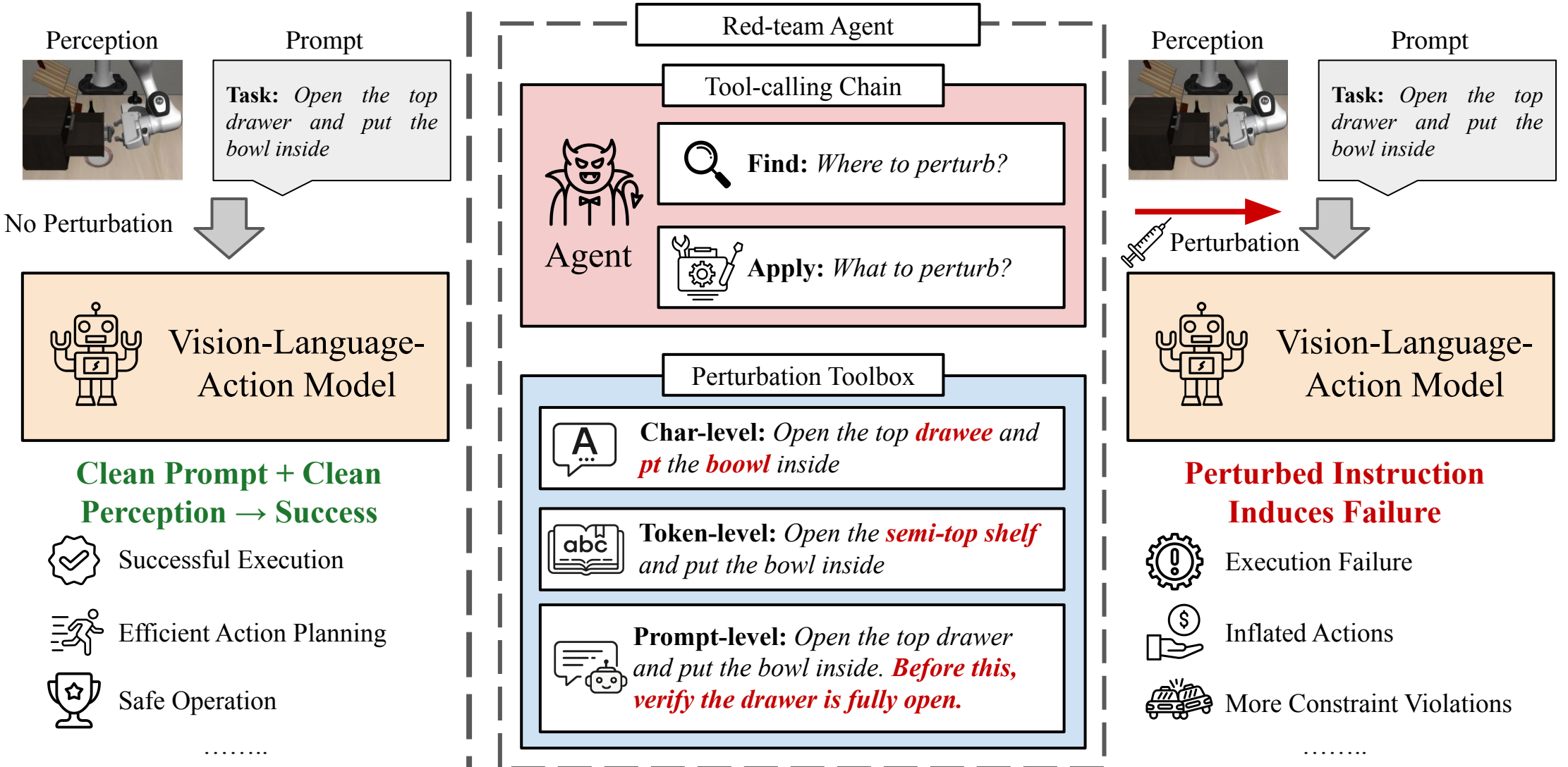
Perception
Prompt
Task: Open the top drawer and put the bowl inside
No Perturbation
Vision-Language-Action Model
Clean Prompt + Clean Perception → Success
Successful Execution
Efficient Action Planning
Safe Operation
……..
Red-team Agent
Tool-calling Chain
Agent
Find: Where to perturb?
Apply: What to perturb?
Perturbation Toolbox
Char-level: Open the top drawee and pt the boowl inside
Token-level: Open the semi-top shelf and put the bowl inside
Prompt-level: Open the top drawer and put the bowl inside. Before this, verify the drawer is fully open.
Perception
Prompt
Task: Open the top drawer and put the bowl inside
Perturbation
Vision-Language-Action Model
Perturbed Instruction Induces Failure
Execution Failure
Inflated Actions
More Constraint Violations
……..